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Abstract

This paper traces the evolution of the Pershing II (PII) missile system, fielded in 1983 along with the Ground-Launched Cruise Missile (GLCM), to modernize NATO's intermediate range nuclear forces (INF), in response to the continuing buildup of Soviet SS-20 missiles and Backfire bombers aimed at Western Europe. PII weapon system hardware, operations, performance, and mission effectiveness are discussed. The PII is an evolutionary weapon system development in response to changing mission requirements and advances in applicable technology.

Nomenclature

AAA	Anti-Aircraft Artillery
AD	Advanced Development
ARS	Automatic Reference System
ARTY/ORD	Artillery/Ordnance
ATBM	Antitactical Ballistic Missile
C ³	Command, Control, and Communication
C ³ I	Command, Control, Communication, and Intelligence
CAS	Combat Alert Status
CEP	Circular Error Probability
CONUS	Continental United States
DC	Development Cost
DCU	Digital Correlator Unit
DDE	Desired Damage Expectancy
DE	Damage Expectancy
DLMS	Digital Land Mass Simulation
DMA	Defense Mapping Agency
DSARC	Defense System Acquisition Review Council
ED	Engineering Development
EL	Erector Launcher
ETR	Eastern Test Range
FORSCOM	Forces Command
FOT	Follow-On Operational Test
FRG	Federal Republic of Germany
G&C/A	Guidance and Control/Adapter
GLCM	Ground-Launched Cruise Missile
INF	Intermediate Range Nuclear Forces
JCS	Joint Chiefs of Staff
LCC	Life Cycle Costs
LRINF	Longer Range Intermediate Range Nuclear Forces
NATO	North Atlantic Treaty Organization
M.A.N.	Maschinenfabrik, Augsburg, Nurnberg AG
ODB	Operational Data Base
O&SC	Operations and Support Costs
PC	Procurement Cost
PCC	Platoon Control Central
PERSEP	Pershing Survivability Evaluation Program
PIa	Pershing Ia
PII	Pershing II
PLS	Prelaunch Survivability
POA	Probability of Arrival
PRESS	Pershing Reference Scene System
PS	Power Station
PSSK	Probability of Single Shot Kill
PTP	Probability to Penetrate
QRA	Quick Reaction Alert
RAC	Radar Area Correlator
RCS	Radar Cross Section
RLCU	Remote Launch Control Unit
RS	Radar Section
RSGF	Reference Scene Generation Facility

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RV	Reentry Vehicle
SAM	Surface-to-Air Missile
SCTS	System Components Test Station
SLA	Sequential Launch Adapter
TNF	Theater Nuclear Forces
TRACE	Terminal Radar Area Correlator Experiment
TV	Target Vulnerability
WHS	Warhead Section
WSR	Weapon System Reliability
WSMR	White Sands Missile Range
WOF	Weather Operational Factor
Y	Yield (Warhead)

Introduction

Recognizing the extensive Soviet modernization and buildup of theater nuclear forces (TNF), particularly mobile SS-20 longer range intermediate range ballistic missiles and Backfire bombers, the NATO nations jointly decided in 1979 that NATO TNF must be modernized to counter this increasing Soviet threat, and that negotiations aimed toward limitation or reduction of these TNF missiles should be initiated.

In a communique issued after a special meeting of the NATO Foreign and Defense Ministers in Brussels on 12 December 1979, the ministers decided to modernize NATO's longer range intermediate nuclear range forces (LRINF) while initiating arms control discussions. A modernization plan having two parallel and complementary paths was implemented. This plan called for the deployment of 108 Pershing II (PII) launchers (to replace existing Pershing Ia launchers), on one hand, and the deployment of 464 Ground-Launched Cruise Missiles (GLCMs) on the other. All these missiles have single warheads. The PII became operational in December 1983.

This paper addresses the PII element of the NATO LRINF modernization. Its evolution, operation, characteristics, and the factors contributing to its effectiveness are discussed.

Background

At the end of World War II, the U.S. and the USSR acquired significant expertise and military hardware originally developed by the Germans. This expertise included state of the art guided missile technology for both cruise and ballistic missiles as exemplified by the FZG-76 (or V-1) and the A-4 (or V-2). Although these weapons represented significant departures from manned bombers used by both sides during World War II for long range weapon delivery, they did not have the terminal accuracy needed to make them effective tactical nonnuclear weapons. Evolution of U.S. cruise and ballistic missile systems since the V-1 and V-2 is shown in Figures 1 and 2.

Robert Lusser, former director of aircraft development at the Gerhard Fiesler Werke in Kassel, Germany and V-1 development authority, made the following recommendations in a lecture in 1949 regarding future surface-to-surface tactical guided missile development. He based his recommendations on German V-1 experience:

- 1 "In the case of the V-1 it has been proven that light mobile launching sites were much safer against counteraction than heavily protected static emplacement."
- 2 "The missiles should be designed carefully and prepared so as to be capable of storage in the open air."

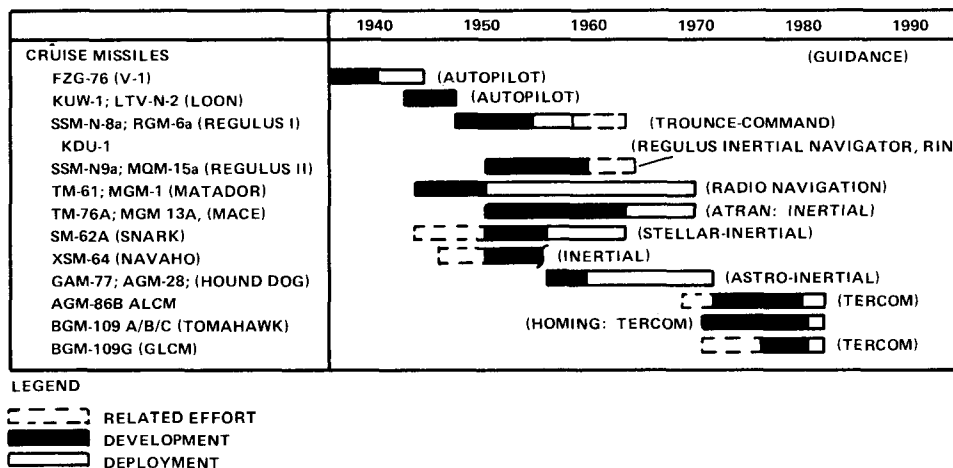


Figure 1. U.S. Cruise Missile Evolution

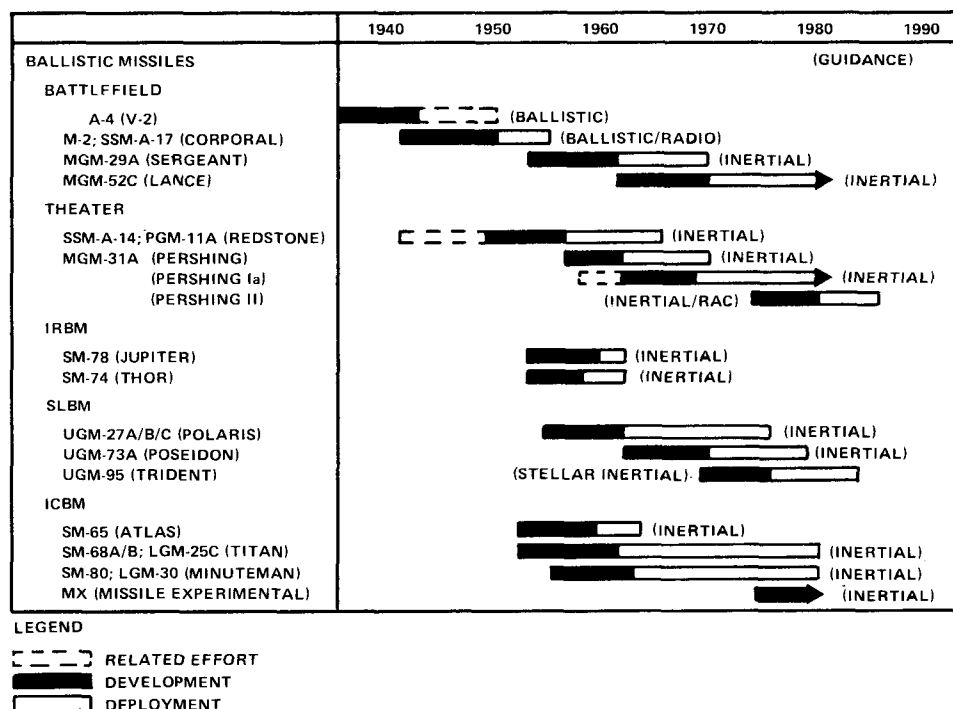


Figure 2. U.S. Ballistic Missile Evolution

3 "The service altitude of a missile should be high enough to avoid small-caliber anti-aircraft gunfire."

4 "The final mass production of a missile should not be initiated before the results of full-scale maneuvers are evaluated and incorporated in the working drawing."

The first two recommendations contributed to prelaunch survivability. Most Soviet and U.S. tactical ballistic missile systems (e.g., SS-20, SS-21, SS-22, SS-23, Lance, and Pershing) are mobile launcher systems. Most of these systems provide for field-based storage and support in operational use.

Considering the significant increases in anti-air-warfare systems, including the many Soviet surface-to-air missile (SAM) systems, the third recommendation pertaining to probability to

penetrate should be extended to include anti-aircraft artillery (AAA) and missile coverage zones. Longer range tactical ballistic missiles, such as Pershing II, overfly these air defense coverage zones, while cruise missiles attempt to underfly these zones. Recent improvements in U.S. and Soviet look-down/shoot-down air defense system capabilities appear to make the underflight of air defenses more difficult, without significantly increasing probability of ground clobber.

The fourth recommendation has led to serial replacement of similar systems by Soviet and U.S. forces (e.g., FROG-7 replaced by SS-21, SCUD by SS-23, SCALEBOARD by SS-22, Honest John and Sergeant by Lance, and Redstone by Pershing). The importance of field experience with operational personnel and logistics also led to increased emphasis on operational evaluation of U.S. systems.

Pershing Evolution

The original Pershing development program was initiated in January 1958. Martin Marietta Aerospace was awarded the prime contract in March 1958. The first Pershing battalion was activated at Fort Sill, Oklahoma in June 1962. Pershing was deployed with U.S. units in Europe in 1964 and German Air Force units in 1965. The original Pershing system was mounted on M-474 (modified M-113) tracked vehicles.

By 1965 Pershing units in Europe were assigned a quick reaction alert (QRA) mission. To increase the system's ability to shoot, move, and communicate in the QRA role, the Pershing Ia (PIa) system was developed and deployed in Europe in 1969. PIa replaced the tracked launcher vehicles with wheeled vehicles and had an improved erector launcher that achieved improved response time. Additional system improvements, including the automatic reference system (ARS) and sequential launch adapter (SLA), were fielded in 1976. The ARS provided automatic alignment of the missile's inertial reference system without pre-surveyed sites, and the SLA allowed countdown and launch of three missiles without moving the launch equipment or cables.

The major stages of Pershing evolution are shown in Figure 3 and listed in Table I.

Pershing II Evolution

As a result of radar area correlation missile guidance studies for AGMX-3 and TIMS in 1972, a program was started to verify the terminal guidance accuracy predicted for a modular improvement being proposed for the Pershing surface-to-surface missile system deployed in Europe. Helicopter and fixed-wing captive flight tests were conducted with breadboard systems. The success of the Terminal Radar Area Correlator Experiment (TRACE) program led to initiation of the Pershing advanced development (AD) program in 1974, following the Defense System Acquisition Review Council I (DSARC).

The PII AD program included six scheduled flights of the modified PIa missile with radar area correlation (RAC) guidance. After five flights, all objectives were met. Thus, the accuracy and reliability of the RAC guidance system were established and the test program terminated. Earlier, extensive captive flight tests using rotor and fixed-wing aircraft of prototype systems over thousands of hours built a strong confidence in the capabilities of the RAC guidance equipment.

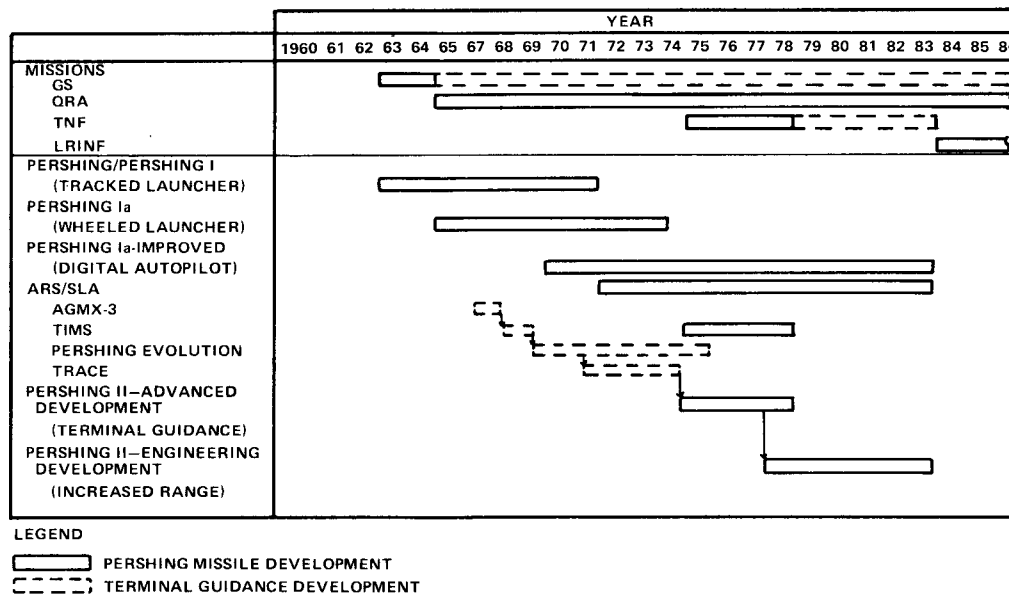


Figure 3. Pershing System/Mission Evolution

Table I. Pershing Evolution

PERSHING I	PERSHING Ia	PERSHING II
1958 - Initial development contract	1965 - PIa development contract	1971 - Terminal guidance IR&D task
1960 - First development firing	1967 - PIa environmental tests - Eglin	1972 - Trace contract
1961 - PI production contract	1967 - PIa production (GSE) contract	- Helicopter flight tests
1962 - First full flight (400 miles) ETR	1968 - Road tests at Orlando	1974 - RADAC contract
- First battalion activated (2/44)	- CONUS SWAP began	- Fixed wing flight tests
- Cold weather tests - Alaska	1969 - PIa deployed in Europe	- DSARC I
1963 - Tropical tests - Panama	1970 - U.S. deployment completed	1975 - PII AD contract
- Tactical firings - WSMR	1971 - GAF deployment completed	1977 - PII AD flight tests - WSMR
1964 - PI deployed in Europe	- Missile and PS development began	- (captive and missile)
- GAF begins PI training - U.S.	1972 - ARS/SLA development began	- (analog correlator)
1965 - PI assigned QRA mission	1973 - Missile and PS improvements fielded	1978 - DSARC II
- PI deployed with GAF	- Missile production reopened	1979 - PII ED contract
1968 - SWAP (PIa replacement) begins	1974 - First firings with ARS/SLA	- Captive flight tests (digital correlator)
1969 - SWAP (begins in Europe)	1975 - ARS/SLA production contract	- NATO LRTNF decision
1970 - U.S. SWAP completed	1976 - ARS/SLA issued to units	1980 - PII motor static firings
1971 - GAF SWAP completed	- Missile production closed	1982 - ED flight tests begin
	1978 - Missile production reopened	- Long lead procurement
	1980 - 400th Pershing missile fired	- ASARC/DSARC III
	1981 - PIa missile production completed	- PII production contract
	1983 -	
	1988 - PIa to PII SWAP scheduled	1983 - Complete ED flight tests
		- IOC Europe

Originally, the PII system was conceived as a “modular” improvement to PIa. It was to use the same motors and ground equipment plus have the same 740 km range as PIa. The only basic difference was the addition of terminal guidance for greater accuracy.

After completing the AD program, however, a decision was made to extend the PII range beyond that of PIa. The extended PII range and its improved accuracy significantly increased NATO’s flexibility and military effectiveness.

In 1979, following DSARC II, a full scale engineering development (ED) program for PII began. At that time, 28 test flights were planned, beginning in April 1982 and ending in late 1983. This test program was reduced in December 1981 to 18 flights by combining several development and operational tests.

To meet the December 1983 deployment required by the 1979 NATO Ministers agreement, it was necessary to authorize production before the first test launch of the ED missile. The ED program was designed to complete the most critical operational ground and captive guidance testing before the first PII flight to minimize risk before the start of production. Additionally, all ED missiles were to be built with the same manufacturing processes and tools that would be used in the production program.

Pershing In Europe

A PIa force in Europe is continually on QRA status. During normal peacetime operations, these units are deployed in specially prepared fixed combat alert status (CAS) sites. During periods of tension or wartime, all deployed Pershing units are moved to field positions that frequently change to increase their prelaunch survivability.

The PII units will conduct peacetime operations and training essentially the way existing PIa units now operate. When on CAS, a battery occupies one of the prepared QRA sites and stands in a maximum state of readiness. The battery will then go on field maneuvers for training. Between this phase and other exercises, a unit will spend a considerable portion of time in the field.

During these periods, the unit is visible to the public as it moves about the countryside. The battery then returns to its kaserne for maintenance and preparation for return to the QRA site.

The U.S. Army, 56th Field Artillery Brigade (Pershing) and German Air Force Pershing units are deployed in West Germany. The 3rd Battalion, 9th Field Artillery, a Forces Command (FORSCOM) unit, is stationed at Fort Sill, Oklahoma with the primary mission of supporting Continental United States (CONUS) Pershing firing programs and Pershing school training programs in Fort Sill. In addition to the PIa units deployed in Europe and CONUS, all required schools, range, and support facilities for the Pershing weapon system are currently operational and will continue to support the PII units.

Pershing Flight Tests

The Artillery/Ordnance (ARTY/ORD) and follow-on operational test (FOT) firing programs have been conducted for the past 10 years with the PIa missile system and previously with the Pershing system. These test crews, missiles, and related support equipment are typically taken intact from an operational field site to Patrick Air Force Base and the Air Force Eastern Test Range (ETR) in Florida for follow-on operational test (FOT), or to White Sands Missile Range (WSMR) in New Mexico for ARTY/ORD tests. The missiles are fired much as they would be in Europe.

The 400th Pershing missile was fired by a German crew at WSMR on 15 October 1980. At that time, the Army Pershing Program Manager, Colonel William J. Fiorentino, noted that the Pershing system had established a commendable record in schedule, cost, and reliability, making it a top contender for the most successful major weapon system developed by the U.S. These ongoing test programs have provided extensive experience in all aspects of Pershing weapon system operations. They have provided for evolutionary improvements in the system hardware, organization, and operational procedures.

The PII ED flight test program was completed 18 September 1983. A summary of these flight test programs is shown in Figure 4.

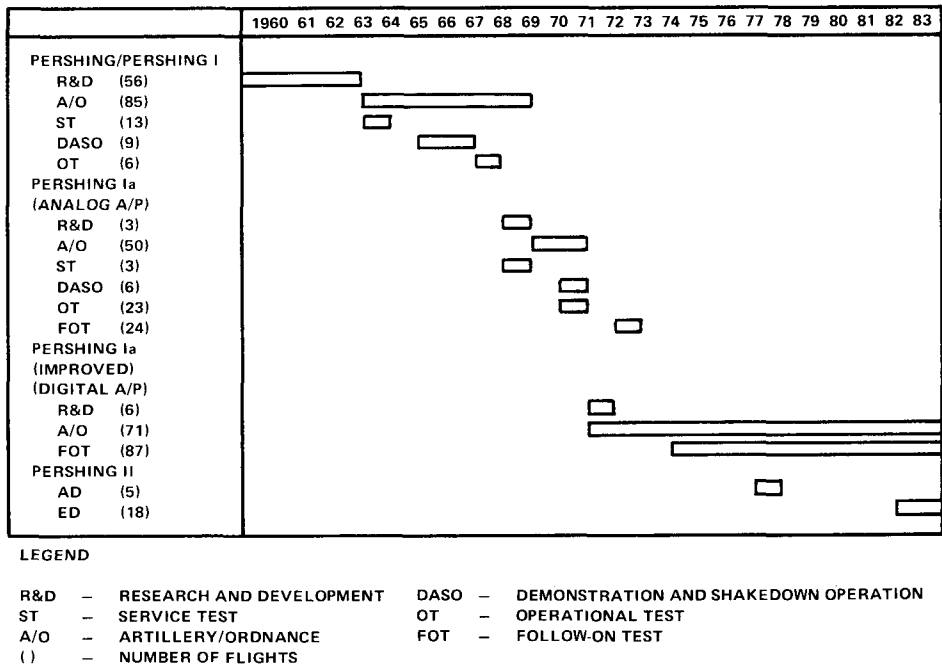


Figure 4. Pershing Flight Test Programs

Pershing II Weapon System

The PII missile represents the current state of the art for tactical ballistic missile systems. It will be the first ballistic missile to be deployed with a terminally guided maneuvering reentry vehicle. It began replacing Pls currently deployed in Europe on a one-for-one basis in December 1983.

Pershing II Hardware Description

Essential hardware for the Pershing system as shown in Figure 5 includes the missile, erector launcher, 10-ton M.A.N. truck-tractor, and the forward area cable set which includes the remote launch control unit (RLCU) containing missile firing controls. Adequate communication equipment ensuring essential command, control, and communication (C³), including nuclear release and weapon system control, are an integral part of the unit.

Normal (three-missile) firing platoon operations also include the platoon control central (PCC) vehicle housing the three RLCUs (one for each missile).

The reference scene generation facility (RSGF) allows the firing unit to generate its own target tape cartridges for all possible targets.

The system components test station (SCTS), rear area power unit, electrical and mechanical shop sets, plus other support equipment are normally assigned to Pershing support battalion units that provide necessary support during long term Pershing system deployments.

Containers and missile slings are provided for transport and assembly of disassembled Pershing missile rounds. Standard

M-127 semitrailers (two for each missile) with appropriate truck tractors would normally be used to transport disassembled missile rounds and warheads.

Pershing II Missile. The PII reentry vehicle (RV) is structurally and functionally divided into three sections: the radar section (RS), warhead section (WHS), and guidance and control/adaptor (G&C/A) section. Quick access splices make the RV sections completely replaceable at the firing site.

The RS consists of the radar unit with the antenna enclosed in an ablative radome. The function of the radar unit is to transmit radio frequency energy to the target area, receive altitude and video return, and route the detected video and altitude data to the digital correlator unit (DCU) located in the G&C/A.

Two major structural parts compose the first and second stages of the PII missile: the motor case with forward and aft stubs and the aft motor skirt. A Kevlar motor case serves as both propellant container and outer skin for the missile in the center of the section. The forward motor stub contains a grooved band to accept the V-band that splices the G&C/A section to the second stage propulsion section. Structural support between the motor case and azimuth ring of the erector launcher is provided by the first stage aft motor skirt. The aft motor skirt also provides support for the swivel-nozzle actuation system, frame shield, vane control system, and tail plug assemblies of the first stage propulsion section. A conduit cover is externally mounted between the forward motor stub and the aft motor skirt. Hard points are provided on the forward stubs and aft skirts for section handling. Two fixed and two movable vanes (for roll control) in an "X" configuration are located on the first stage aft skirt. No control surfaces are required on the second stage propulsion section.

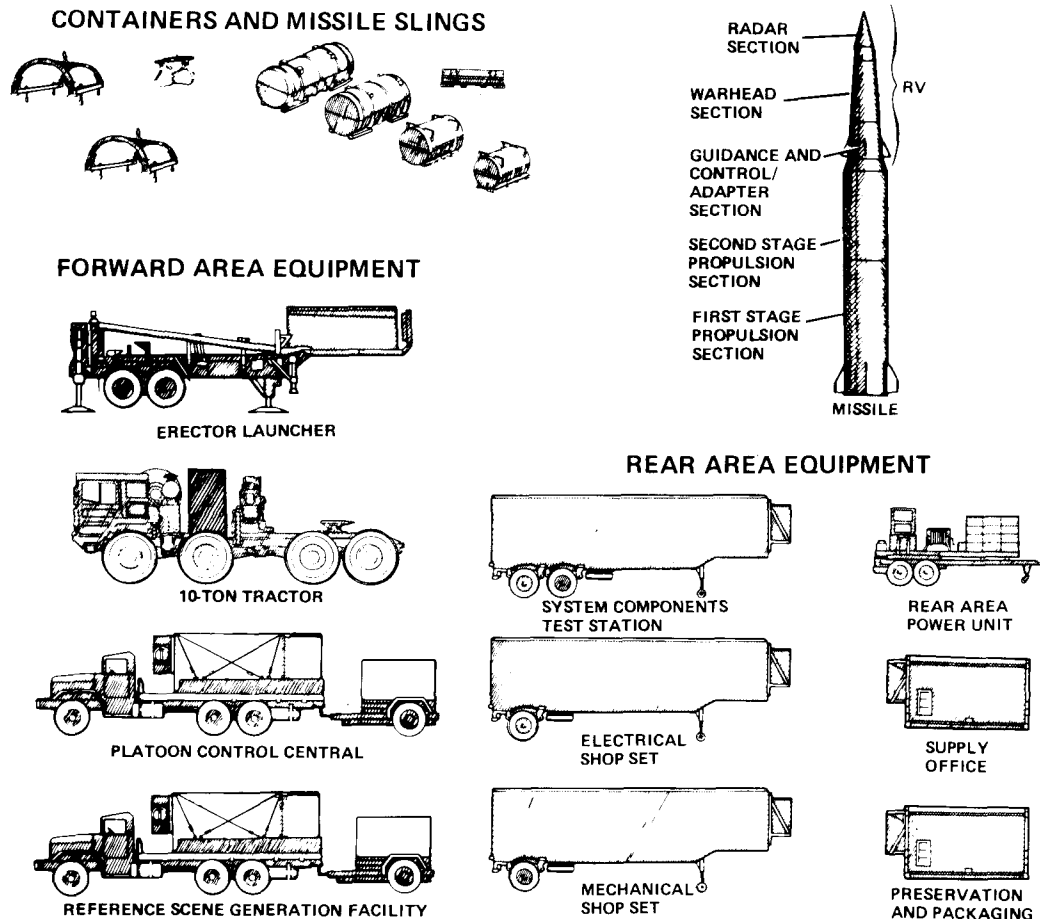


Figure 5. Pershing II Major Equipment

Pershing Reference Scene System. The Pershing reference scene system (PRESS) converts the digital land mass system (DLMS) feature and elevation data at the Defense Mapping Agency (DMA) into the format required to generate PII target reference scenes. The output of the PRESS can be disk packs containing the PII operational data base (ODB) for all targets or discrete radar reference scene cartridges as shown in Figure 6.

Planned operations may be accomplished using reference scene cassettes shipped to the user in Europe. Using the ODB required target, reference scenes may also be generated at the user's battalion level.

As changes to the DLMS data base are made, the PRESS will generate updated ODB disks that would be forwarded to the user. From the new data base, platoon reference scene cartridges can be updated.

Pershing II Operations

Normal peacetime Pershing operations in Europe are conducted by a three-platoon firing battery of each battalion on QRA covering assigned targets, from a permanent hardsite.

When appropriate, alert measures are ordered, previously unused field firing sites are occupied, and responsibility for coverage of all assigned targets is assumed. Once fully deployed, the firing elements are deployed by platoon (Figure 7) and geographically separated to reduce vulnerability. These firing units are moved to new firing positions on a random schedule to increase survivability.

The capability exists for an independent countdown and launch of a single PII missile by an erector launcher dispersed from its platoon. The only launch equipment required for the launcher is the RLCU. It can be removed from the PCC for this purpose. Radios for communications with the parent unit also accompany the launcher along with release message description and authentication capabilities, if required. This optional independent launcher deployment mode allows maximum dispersion of even a small PII unit, thereby significantly enhancing prelaunch survivability.

Pershing II Performance

Prior to launch, PII is referenced in azimuth by its gyrocompassing inertial platform. Launch site location is obtained by inspection from large scale DMA maps.

Magnetic tape cassettes containing target reference scenes for the RAC can be provided by DMA using PRESS or prepared by the firing unit using the RSGF.

After launch, the missile follows an inertially guided trajectory until RV separation. Attitude and guidance commands during powered flight (except for roll attitude) are executed by the swivel nozzles in both propulsion sections. Roll control is provided by two movable air vanes on the first stage motor during first stage flight. The first stage also has two fixed air vanes for stability during first stage powered flight.

The midcourse phase of the trajectory (Figure 8) is initiated by RV separation and continues until the terminal phase begins. At the beginning of the midcourse phase, the RV is pitched down to orient it for atmospheric reentry and to reduce its radar cross section (RCS). Midcourse attitude is controlled by the RV vane control system during atmospheric exit and reentry. In addition, a reaction control system is used during exoatmospheric flight.

At a predetermined altitude above the target, the terminal phase begins. A velocity control maneuver (pullup/pulldown) is executed under inertial guidance control to slow the RV and achieve the proper terminal velocity. The radar correlator system is activated, and the radar scans the target area. Radar return data

are compared to prestored target reference data. The resulting position fix information is used to update the inertial guidance system and generate RV steering commands. The RV is maneuvered to the target by the RV vane control system. The correlation process is performed several times and continues to update the inertial guidance system until just prior to impact (Figure 9).

If for some reason the correlator system fails to operate or if the correlator data quality is found to be faulty, the inertial guidance system continues to operate and guides the RV to the target area with inertial accuracy. Inertial drift errors are minimized in PII due to its short time of flight.

The improved PII accuracy is also effective against hard targets that might not be adequately attacked by currently available weapon systems, including PIA.

Pershing II Mission Effectiveness

President Ronald Reagan said he believes actual deployment of PII missiles in Europe is necessary to involve the Soviet Union in "legitimate" INF arms control negotiations. The primary mission for a modernized NATO INF composed of PII missiles and Ground-Launched Cruise Missiles (GLCMs) is to deter Soviet/Warsaw Pact aggression against NATO. More specifically, it is to deter Soviet longer range SS-20 and Backfire weapons, or the threat of their use against NATO.

There are no generally accepted analytical models for evaluating the effectiveness of a weapon system as a deterrent. There are, however, generally accepted factors used to assess the cost effectiveness of tactical weapon systems. Some primary factors considered for comparative cost effectiveness of candidate LRINF elements are shown in Figure 10. These factors may be roughly categorized as mission related factors, weapon system factors, force structure/deployment factors, and cost factors.

The measure of effectiveness normally used for an INF system is the expected number of high priority target kills for a given force size. A desired damage expectancy (DDE) is specified for all tactical targets, and if less than the DDE is predicted for a specific target, then multiple rounds are used to achieve the DDE. NATO planners have specified DDE planning factors for particular classes of high value targets.

The DDE for a specific target is the measure of salvo kill probability (i.e., the cumulative probability of kill of a salvo of n weapons, each with a given damage expectancy (DE).

$$DDE = 1 - (1 - DE)^n \quad (1)$$

DE is the product of single shot kill probability (PSSK) and probability of arrival (POA).

$$DE = PSSK \times POA \quad (2)$$

PSSK is a function of target vulnerability (TV), warhead yield (Y), damage causing effects and delivery accuracy, or circular error probability (CEP).

$$PSSK = f(TV, Y, CEP) \quad (3)$$

POA is the probability that a missile (weapon system) will survive enemy action, is available, and performs reliably through fuzing and warhead detonation against a specific target. It is the product of prelaunch survivability (PLS), weapon system reliability (WSR), probability to penetrate (PTP) enemy defenses, and a weather operational factor (WOF) which accounts for target obscuration due to weather (primarily in visual or optical sensor systems) and other significant natural/neutral environment effects such as low altitude missile clobber probability.

$$POA = PLS \times WSR \times PTP \times WOF \quad (4)$$

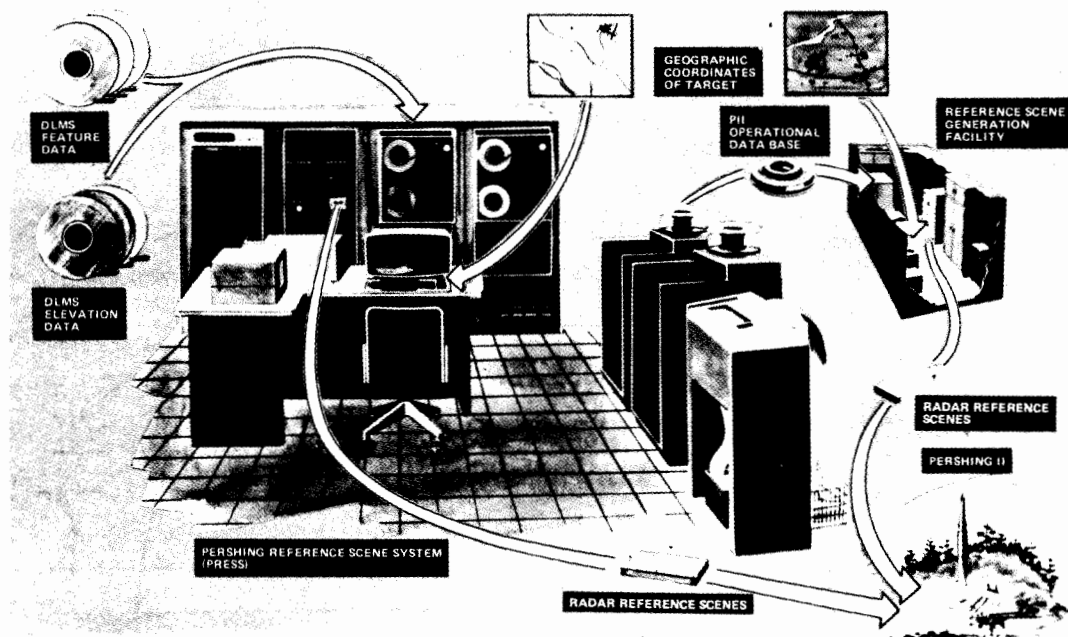


Figure 6. Pershing Reference Scene System

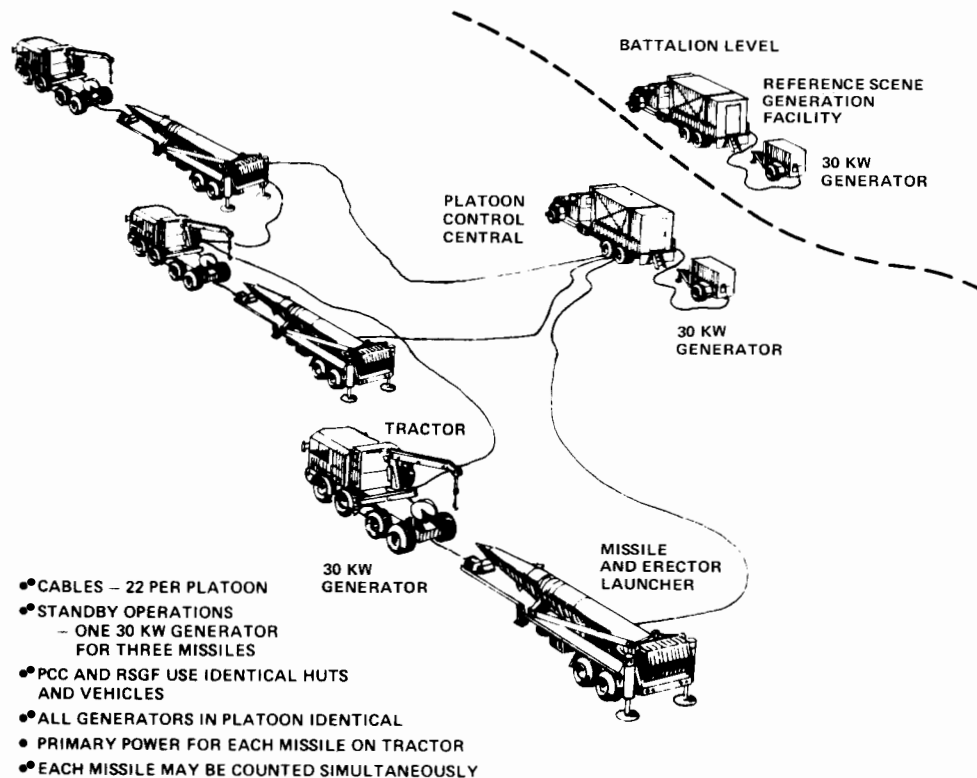


Figure 7. Pershing II Firing Platoon Hardware

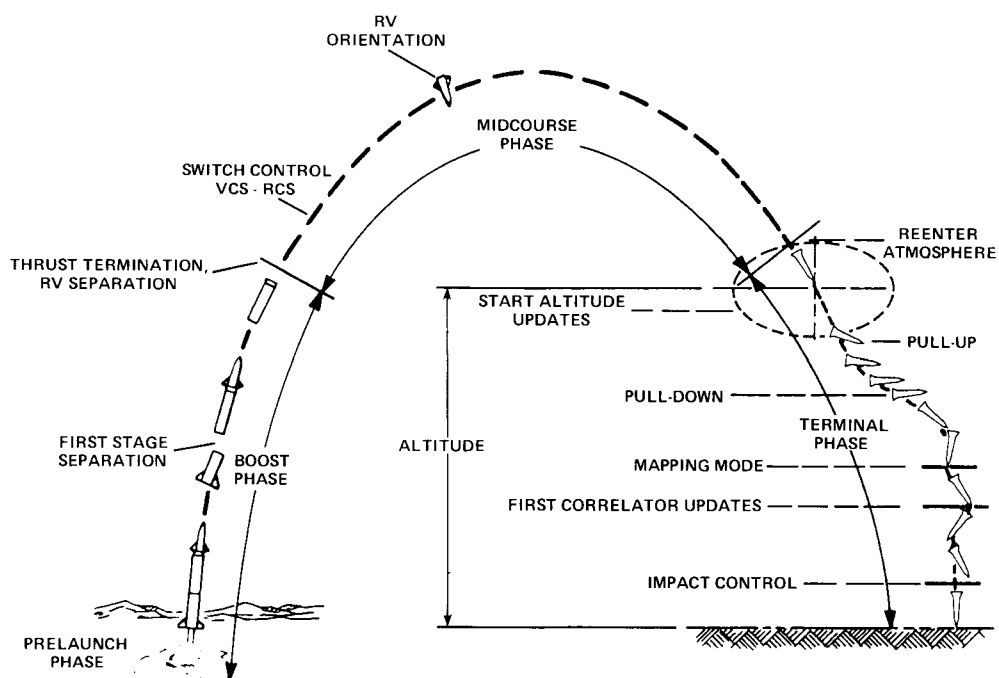


Figure 8. Pershing II Trajectory

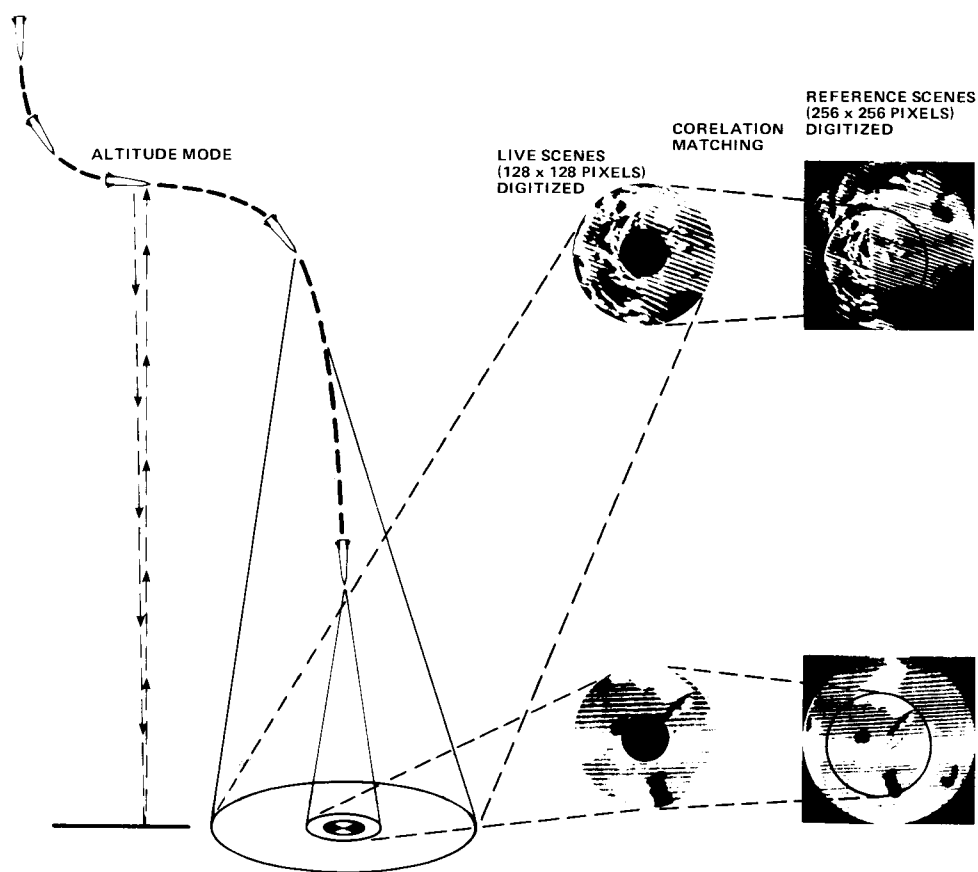
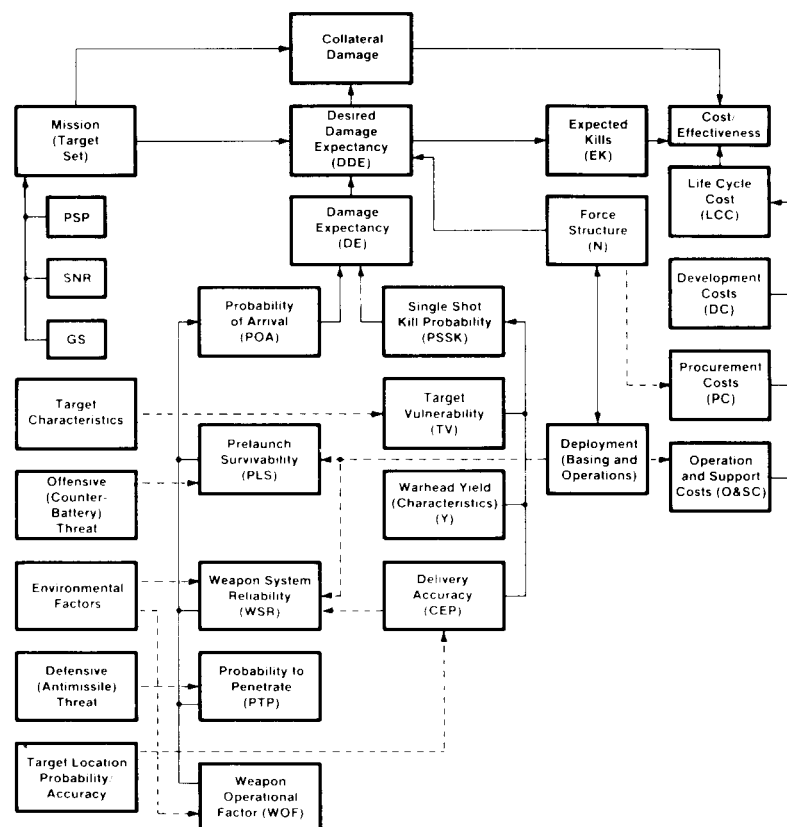


Figure 9. Radar Unit In-Flight Operation



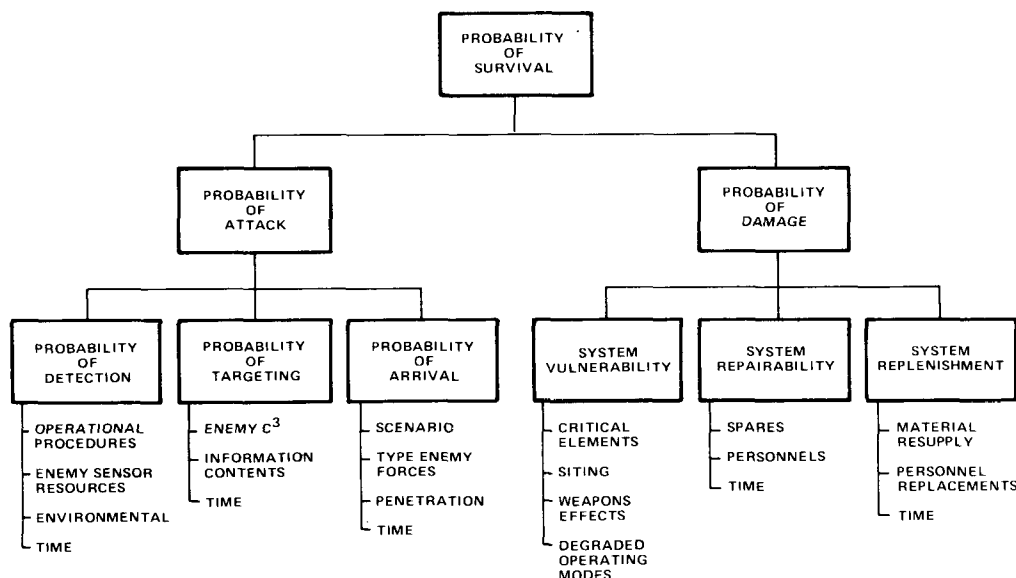


Figure 11. Prelaunch Survivability Factors

Weapon System Reliability (WSR). WSR estimates for all strategic nuclear weapon systems are required by the Joint Chiefs of Staff (JCS) for use in nuclear planning. The validity of these estimates for deployed systems is verified by periodic missile firings, including Pershing follow-on operational test (FOT) firings which are performed by randomly selected QRA personnel. These tests are designed not only to test the reliability and accuracy of the deployed hardware, but the entire system (i.e., personnel, logistics, training, command, control, communications, and other support systems).

Reliability of missile flight systems is a complex function of component quality, mission time, performance functions, and the operational environments that subject components to various stresses.

Weather Operational Factor (WOF). The PII missile, because of its high partially exoatmospheric ballistic trajectory, is not affected by enroute weather conditions, obstacles, and terrain.

Probability of Kill (Pk). PII can be used with inertial only (degraded accuracy) or inertial with terminal guidance (full accuracy). Ballistic flight profiles provide high kinetic energy and verticality which contribute to the effective delivery of many conventional munitions and earth penetrator warheads.

Force Structural/Deployment Factors

The NATO INF modernization plan calls for replacement of the 108 deployed U.S. Pla launchers with 108 PII launchers. In addition to the Pershing units currently deployed in Europe and CONUS, all of the required schools, range, and support facilities for the Pershing weapon system exist and are currently operational. The existing Pla support activities will continue for PII.

Dispersion mobility can be costly in terms of force structure. The 20 years of Pershing field experience in Europe has led to a cost effective balance between the increase in PLS and cost of dispersion. Also, because of this experience, operations and support costs (O&SC) models developed for the Pershing weapon system are recognized among the best available.

Cost Factors

The PII system O&SC represent a refinement to costs for a system that has been deployed in NATO since 1964. PII costs may be too well defined to objectively compare with a new weapon system not yet deployed in the extremely complex multinational environment. PII launchers are modified versions of existing Pla equipment. No new support facilities will be required.

Summary and Observations

The Pershing II missile system was initially conceived as the incorporation of radar area correlation (RAC) terminal guidance into the deployed Pershing Ia system to minimize collateral damage. After the successful development and demonstration of RAC guidance (analog) and a nuclear earth penetrator warhead in the PII advanced development program after only five missile flights, operational requirements for the PII system were significantly changed.

During the PII ED program, digital RAC terminal guidance replaced the original analog guidance. This significantly simplified operational targeting and support requirements, plus improved inherent system reliability. It provided improved operational performance (more than doubling maximum range and reducing delivery error by an order of magnitude) and incorporated state of the art technology (in propulsion and digital RAC terminal guidance) which improved not only performance, but also system reliability.

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